

Haider Riaz Khan

🏠 haiderriazkhan.com ✉ haiderriazkhan@gmail.com
in haider-riaz-57593aba 🌐 haiderriazkhan

EDUCATION

2024 – 2026	M.S. in Mathematics	CITY COLLEGE OF NEW YORK
2017 – 2018	M.A. in Philosophy Thesis: The Phenomenological Origins of Property (Advised by John Turri)	UNIVERSITY OF WATERLOO
2010 – 2015	B.Sc. in Physics and Computer Science	MCGILL UNIVERSITY

ENGINEERING EXPERIENCE

2025 - Pres	Co-Director and Lead Engineer	RED CALENDAR (NYC, USA)
	Lead engineering across projects, beginning with Red Calendar, an independent NYC arts and action calendar, and a custom events platform for Wesleyan Center for the Arts. Built hybrid lexical and semantic search over event data, submission and editorial workflows, authentication, syndication, analytics, and automated deployment infrastructure.	
2024 - Pres	Software Engineer	BASE EDUCATION (NYC, USA)
	Wrote GraphQL APIs and built out the application logic for delivering data insights—including computing student-level data, aggregate statistics, and advanced features such as differentially private statistics. Created fast, efficient, and secure data pipelines that power the backend systems.	
2022 - 2023	Lead Software Engineer	TERRACE (NYC, USA)
	Integrated multiple centralized digital exchanges into Terrace's liquidity platform using a FIX API solution. Worked with Circle REST API suites to enable global payments, withdrawals, and trades using USDC. Interviewed, hired, and mentored junior engineers.	
2021 - 2022	Senior Software Engineer	AVENUE 8 (NYC, USA)
	Implemented Avenue 8's observability infrastructure. Wrote automation, monitoring, and alerting scripts. Automated the management of AWS resources using Terraform. Improved network security.	
2019 - 2021	Software Engineer	LIFION BY ADP (NYC, USA)
	Built Lifion's graph database framework using a microservices architecture, Node.js, Gremlin, and distributed message queuing services. Wrote optimized Gremlin queries and SQL subroutines.	
2016 - 2017	Software Engineer	MEDIDATA SOLUTIONS (NYC, USA)
	Contributed to the development of a Java-based combinatorial test data generator called BIT-TAG.	

AWARDS & HONORS

2025	Dr. Barnett and Jean Hollander Rich Graduate Scholarship	CITY COLLEGE
2025	Dr. Barnett and Jean Hollander Rich Summer Fellowship	CITY COLLEGE
2018	Paul Seligman Memorial Scholarship	UNIVERSITY OF WATERLOO
2015	Dean's Multidisciplinary Undergraduate Research List	MCGILL
2014	NSERC-CREATE Neuroengineering Award	MCGILL
2010	Alexander Rutherford Scholarship	WESTERN CANADA HIGH SCHOOL

PUBLICATIONS

CONFERENCE AND JOURNAL ARTICLES

1. **Khan, Haider Riaz**, & John Turri. "Phenomenological Origins of Psychological Ownership." *Review of General Psychology* 26, no. 4 (December 2022): 446–63.

2. Li, Nan, Yu Lei, **Haider R. Khan**, Jingshu Liu, and Yun Guo. "Applying Combinatorial Test Data Generation to Big Data Applications." In *Proceedings of the 31st IEEE/ACM International Conference on Automated Software Engineering (ASE '16)*, 637-47. New York, NY: Association for Computing Machinery, August 2016.

ESSAYS

1. "The Forever War." *Commune*, Issue 5, Winter 2020.

RESEARCH & TEACHING

2018	Translator	PHILOSOPHICAL SCIENCE LAB, UNIVERSITY OF WATERLOO
	Translated English stimuli into Pashto and Urdu for the purposes of a cross-cultural study of the "ought implies can" principle. Provided feedback on research paper drafts to Prof. Turri.	
Winter 2018	Teaching Assistant	UNIVERSITY OF WATERLOO
	Biomedical Ethics (Prof. Andrew Stumpf)	
Fall 2017	Teaching Assistant	UNIVERSITY OF WATERLOO
	Professional and Business Ethics (Dr. Jim Jordan)	
2014 - 2015	NSERC Neuroengineering Fellow	MONTREAL NEUROLOGICAL INSTITUTE
	Wrote CANDLE-J, an open-source 3D image denoising software designed as an ImageJ plugin. CANDLE-J is adept at processing deep in vivo 3D multiphoton microscopy images where the signal-to-noise ratio (SNR) is low. It is written in Python, Java, and multithreaded C.	
2012 - 2014	Computational Scientist	COOK LAB, MCGILL UNIVERSITY
	Developed MATLAB routines to measure the cross-correlation of microsaccades and microstimulations (in area MT of the visual cortex). Created a library of MATLAB functions to compute joint metrics of neural activity. The joint metrics are computed for both single-unit and multi-unit neuronal spikes.	
Fall 2011	SUS Peer Tutor	MCGILL UNIVERSITY
	Met with students routinely to go over physics and chemistry concepts; held exam review sessions.	

TALKS & PRESENTATIONS

2016	comBinatorial bIg daTa Test dAta Generator (BIT-TAG)	ASE'16, SINGAPORE
2014	Neuroscience From Below	STUDENT SUMMER COLLOQUIUM, MCGILL

TECHNICAL SKILLS

Programming Languages: JavaScript/TypeScript, Clojure, Java, Python, C/C++, Bash, SQL
Data, Storage & Querying: MySQL/Postgres, pgvector, S3, GraphQL, Elasticsearch, Redis
Development & Cloud: Git, GitHub Actions, Docker, AWS Lambda
Scientific Computing: MATLAB, Jupyter, SageMath, ImageJ, dtype-next

LANGUAGES

English, Urdu, Pashto